

BLACKBOX X — PHASE 3.9

REGIME-AWARE PROBABILISTIC INTELLIGENCE ARCHITECTURE
— Complete Mathematical, Algorithmic, and Technical
Specification — 1.1 Core Research Objective — 1.2 Non-Predictive
Charter & Conceptual Delineation

DOCUMENT

**REGIME-PROBABILISTIC-
ARCHITECTURE.pdf**

SYSTEM

BLACKBOX X Engine

SOURCE

**docs/REGIME-PROBABILISTIC-
ARCHITECTURE.md**

STATUS

Documentation

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BLACKBOX X — PHASE 3.9

REGIME-AWARE PROBABILISTIC INTELLIGENCE ARCHITECTURE

Complete Mathematical, Algorithmic, and Technical Specification

1. Purpose & Philosophical Framing

1.1 Core Research Objective

Phase 3.9 extends BLACKBOX X from unconditional Monte Carlo simulation into **regime-conditioned probabilistic research**.

The primary quantitative research question answered by this engine is:

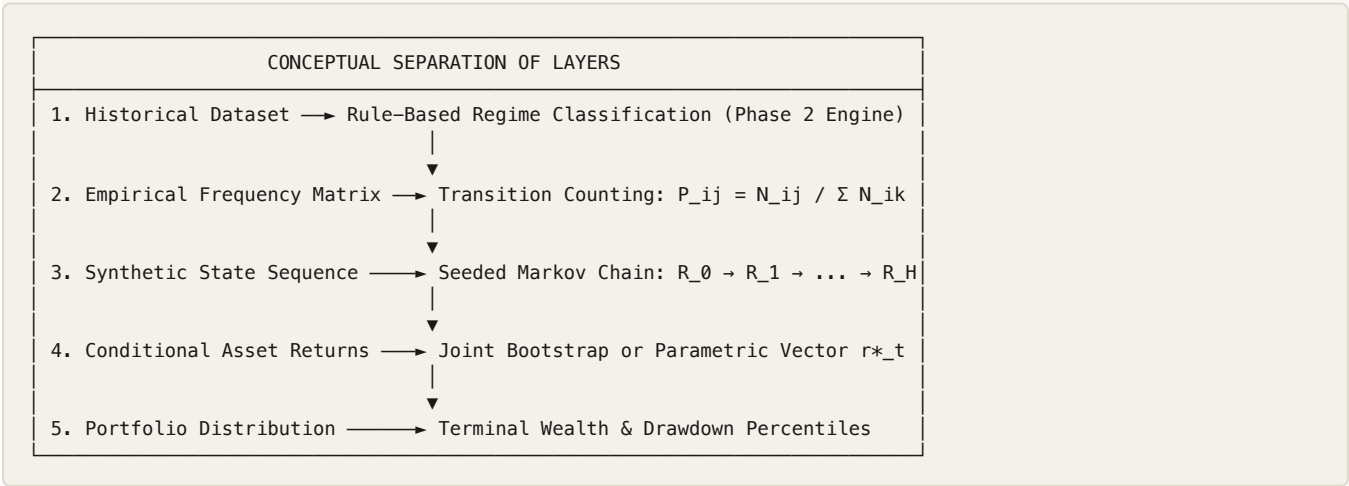
"How does the simulated distribution of portfolio outcomes change when the simulation is conditioned on observed market regimes and empirically observed regime transitions?"

1.2 Non-Predictive Charter & Conceptual Delineation

BLACKBOX X **does not predict future market regimes**.

The engine explicitly separates and maintains four distinct analytical concepts:

- 1. **Historical Observation:** Rule-based regime classifications applied to historical daily price bars in the offline demonstration dataset.
- 2. **Empirical Transition Estimation:** The observed transition counts N_{ij} and frequencies P_{ij} between consecutive daily historical regime states.
- 3. **Simulated Regime Path:** A stochastic sequence of simulated regime states $R_0 \rightarrow R_1 \rightarrow \dots \rightarrow R_H$ generated by a discrete-time Markov chain governed strictly by empirical transition frequencies.
- 4. **Simulated Portfolio Outcome:** The resulting wealth trajectory and drawdown distribution generated by sampling returns conditioned on the simulated regime path.



Required Institutional Framing:

- "BLACKBOX X evaluates how portfolio outcome distributions differ under regime-conditioned simulation assumptions."
- **Forbidden Phrases:**
 - "the market will enter..."
 - "the next regime will be..."
 - "probability of the next crash..."
 - "forecasted regime"
 - "predictive market transition"

2. Existing Architecture Reuse & Integration Boundaries

Phase 3.9 directly reuses the validated engines of Phases 1 through 3.8 without modifying or duplicating underlying code.

BLACKBOX X PLATFORM	
Phase 1 & 2:	Deterministic Data & Metrics Engine (1,825 synchronized days)
Phase 2:	Rule-Based Market Regime Engine (BULL, BEAR, HIGH_VOL, LOW_VOL)
Phase 3.1:	Research Trail & Ghost Mode Insight Engine
Phase 3.2:	Stress Lab (Deterministic Macro Shock Overlay)
Phase 3.3:	Strategy Genome (Topology & Phenotype Graph)
Phase 3.4:	Risk Committee (ResearchPack & Briefings)
Phase 3.5:	AI Research Assistant (Featherless Inference & Tool Sandbox)
Phase 3.6:	Autonomous Quant Research Agent (Hypothesis & Contradiction)
Phase 3.7:	Portfolio Covariance, Simplex Optimizer, Euler Risk, Backtest
Phase 3.8:	Monte Carlo Engine, Mulberry32 PRNG, Web Worker, VaR/CVaR
PHASE 3.9:	REGIME-AWARE PROBABILISTIC INTELLIGENCE (NEW SPECIFICATION)
	– Empirical Regime Transition Matrix
	– Regime-Conditioned Historical Vector Bootstrap
	– Regime-Conditioned Parametric Cholesky Simulation
	– Starting Regime Conditioning (BULL, BEAR, HIGH_VOL, LOW_VOL)
	– Regime Dwell Time & Transition Path Diagnostics
	– Conditional Euler Risk Decomposition

2.1 Reused Components (Strict Zero Duplication)

- 1. Regime Classification Engine (`src/core/regimes.ts`):**
 - Reuses `detectRegimes(prices, 60)` and existing regime definitions (`BULL` , `BEAR` , `HIGH_VOL` , `LOW_VOL`).
 - Zero redefinition of classification thresholds.
- 2. Synchronized Dataset & Covariance Engine (`src/core/portfolio/covariance.ts`):**
 - The 1,825 synchronized daily observations vector across `GOLD` , `BTC` , and `NVDA` .
- 3. Deterministic PRNG Engine (`src/core/portfolio/prng.ts`):**
 - 32-bit `Mulberry32` generator and Box-Muller Gaussian transform.
- 4. Cholesky Factorization & Regularization (`src/core/portfolio/cholesky.ts`):**
 - Reactive Cholesky decomposition $\Sigma_r = \mathbf{L}_r \mathbf{L}_r^T$ with conditional Tikhonov diagonal regularization.
- 5. Portfolio Path Evolution & Bankruptcy Clamping (`src/core/portfolio/monteCarloEngine.ts`):**
 - Buy & Hold organic drift vs next-bar rebalancing, 10 bps friction, and non-negative capital clamping ($r^* \leq -100\% \rightarrow V_{i,\tau} = 0$).
- 6. Web Worker Performance Architecture (`src/core/portfolio/monteCarlo.worker.ts`):**
 - Off-thread non-blocking execution maintaining responsive UI thread.

3. Market Regime Model & Classification Source

3.1 Existing Regime Taxonomy

The platform defines four discrete macro regimes R :

1. **BULL**: Trending positive growth environment (rolling 60-day annualized return $> 5.0\%$, rolling annualized volatility $\leq 35.0\%$).
2. **BEAR**: Sustained negative or stagnant growth (rolling 60-day annualized return $\leq 5.0\%$, rolling annualized volatility $\leq 35.0\%$).
3. **HIGH_VOL**: Elevated volatility regime (rolling 60-day annualized volatility $> 35.0\%$, supersedes return direction).
4. **LOW_VOL**: Subdued volatility regime (rolling 60-day annualized volatility $< 15.0\%$).

3.2 Classification Source & Invariants

- **Source of Truth**: Daily regime labels are determined by evaluating the synchronized portfolio equity trajectory using `detectRegimes(prices, 60)` from Phase 2 / Phase 3.7 (`src/core/portfolio/portfolioRegimes.ts`).
- **Data Period**: Spans the 1,825 synchronized daily return intervals ($t \in [1, 1825]$) from `2019-01-01` to `2023-12-31`.
- **Classification Status**: Labels are strictly **historical post-hoc classifications** of the offline demonstration dataset. They are **never** predictive indicators or forward forecasts.

4. Regime Observation Matrix & Chronological Alignment

4.1 Chronological Sequence Alignment

Let $t \in \{1, \dots, T\}$ be the index of synchronized daily observations ($T = 1,825$). Each observation row contains:

$$\mathbf{O}_t = (\text{date}_t, R_t, \mathbf{r}_t, R_{p,t})$$

where: - date_t : Calendar date in YYYY-MM-DD format. - $R_t \in \mathbf{R}$: Classified regime label at day t . - $\mathbf{r}_t = [r_{\text{GOLD},t}, r_{\text{BTC},t}, r_{\text{NVDA},t}]^T \in \mathbf{R}^3$: Synchronized asset return vector. - $R_{p,t} = \mathbf{w}^T \mathbf{r}_t$: Realized portfolio return.

4.2 Data Integrity Rules

1. **Zero Forward-Fill**: Missing dates or skipped sessions are strictly prohibited.
2. **Zero Synthetic Label Substitution**: Regime labels trace directly to empirical rolling calculations; no synthetic or smoothed labels may be injected.
3. **Strict Contemporaneous Intersection**: Every observation contains authentic contemporaneous data across all three assets and the corresponding regime state.

5. Empirical Regime Transition Matrix

5.1 Transition Counting & Frequency Formulation

For all consecutive observation pairs (R_t, R_{t+1}) where $t \in \{1, \dots, T-1\}$:

1. **Transition Count Matrix** $\mathbf{N} \in \mathbf{N}^{4 \times 4}$:

$$N_{ij} = \sum_{t=1}^{T-1} \mathbb{I}(R_t = i \wedge R_{t+1} = j) \quad \forall i, j \in \mathcal{R}$$

2. Total Outgoing Transitions:

$$N_i = \sum_{j \in \mathcal{R}} N_{ij}$$

3. Empirical Transition Probability Matrix $\mathbf{P} \in [0, 1]^{4 \times 4}$:

$$P_{ij} = \begin{cases} \frac{N_{ij}}{N_i}, & N_i > 0 \\ 0, & N_i = 0 \end{cases}$$

5.2 Row-Stochastic Invariant

For every originating regime i where $N_i > 0$:

$$\sum_{j \in \mathcal{R}} P_{ij} = 1.0$$

EMPIRICAL TRANSITION MATRIX TOPOLOGY				
Originating (i)	► BULL	► BEAR	► HIGH_VOL	► LOW_VOL
BULL	P_BULL,BULL	P_BULL,BEAR	P_BULL,HVOL	P_BULL,LVOL
BEAR	P_BEAR,BULL	P_BEAR,BEAR	P_BEAR,HVOL	P_BEAR,LVOL
HIGH_VOL	P_HVOL,BULL	P_HVOL,BEAR	P_HVOL,HVOL	P_HVOL,LVOL
LOW_VOL	P_LVOL,BULL	P_LVOL,BEAR	P_LVOL,HVOL	P_LVOL,LVOL

Mandatory Disclaimer: The transition matrix $\mathbf{P}$ reflects empirical transition frequencies observed in the offline simulated demonstration dataset. It does not represent an external stochastic prediction of real-world financial transitions.

6. Zero-Data & Sparse Regime Policy

6.1 Thresholds & Failure Handling

To prevent statistical distortion and numerical instability:

- **Minimum Observation Threshold:**

$$\text{MIN_REGIME_OBSERVATIONS} = 20$$

If any regime i has $T_i < 20$ historical observations, the UI and reports display a prominent institutional warning badge:
 ``SPARSE REGIME: Limited observation count (n < 20). Empirical parameters may exhibit elevated sample variance.``

6.2 Zero-Transition Absorption Prevention

If an originating regime i has $N_i = 0$ outgoing transitions:

1. The regime transition distribution is marked as `'INSUFFICIENT_TRANSITION_DATA'`.
2. The simulation engine is **strictly blocked** from silently defaulting or fabricating probabilities.
3. The UI renders the transition row as unavailable (`N/A`), and AI tools return an explicit validation error:
`"Cannot execute regime-conditioned Monte Carlo: Regime [i] has zero observed outgoing transitions."`
4. **Prohibition of Arbitrary Smoothing:** Laplace smoothing (+1 pseudocounts) or arbitrary Gaussian blurring is **strictly forbidden** unless explicitly requested by the user and documented as an artificial smoothing policy.

7. Regime-Conditioned Return Distributions & Joint Vectors

7.1 Conditional Data Partitioning

For each regime $r \in \mathcal{R}$:

$$D_r = \{\mathbf{r}_t \mid t \in \{1, \dots, T\} \wedge R_t = r\}$$

Let $T_r = |D_r|$ be the sample size of regime r .

7.2 Conditional Moment Estimation

1. **Conditional Mean Return Vector** $\boldsymbol{\mu}_r \in \mathbb{R}^3$:

$$\boldsymbol{\mu}_r = \frac{1}{T_r} \sum_{t \in D_r} \mathbf{r}_t$$

2. **Conditional Daily Covariance Matrix** $\boldsymbol{\Sigma}_r \in \mathbb{R}^{3 \times 3}$:

$$\boldsymbol{\Sigma}_r = \frac{1}{T_r - 1} \sum_{t \in D_r} (\mathbf{r}_t - \boldsymbol{\mu}_r)(\mathbf{r}_t - \boldsymbol{\mu}_r)^T$$

7.3 Multi-Asset Joint Vector Preservation Invariant

- **Zero Asset Decoupling:** For every day $t \in D_r$, the return vector $\{\text{BTC}_t, \text{NVDA}_t\}^T = [\text{GOLD}_t, \text{BTC}_t, \text{NVDA}_t]^T$ is treated as an indivisible atomic tuple.
- **Rationale:** Asset decoupling would destroy cross-asset flight-to-safety dynamics (e.g. Gold outperformance during Bitcoin/equity selloffs during `HIGH_VOL` or `BEAR` regimes).

8. Simulation Method A: Regime-Conditioned Bootstrap (`REGIME_BOOTSTRAP`)

8.1 Algorithmic Execution

For each simulated path $m \in \{1, \dots, M\}$:

1. **Initialize State:** Set day $\tau = 0$, wealth $V_{m,0} = V_0$, regime $R_{m,0} = R_{\text{start}}$.
2. **Forward Step Loop** for $\tau = 1 \dots H$:
 - **Step 2a (Return Sampling):** Draw a discrete uniform random index k from the subset index set of the active regime $R_{m,\tau-1}$:

$$k \sim \mathcal{U}\{1, T_{R_{m,\tau-1}}\}$$

Select the synchronized joint vector:

$$\mathbf{r}_{m,\tau}^* = \mathbf{r}_k \in D_{R_{m,\tau-1}}$$

- **Step 2b (Portfolio Update):** Update position values, apply bankruptcy clamping ($r^* \leq -100\% \rightarrow V_{i,\tau} = 0$), compute portfolio return $R_{p,m,\tau}$, and execute rebalancing if triggered.
- **Step 2c (Regime Transition):** Sample the next regime state $R_{m,\tau}$ according to the cumulative transition probabilities of the current row $\mathbf{P}_{R_{m,\tau-1},\cdot}$: Draw $u \sim U[0, 1)$ from PRNG. Find regime j such that $\sum_{l=1}^{j-1} P_{i,l} \leq u < \sum_{l=1}^j P_{i,l}$. Set $R_{m,\tau} = j$.

9. Simulation Method B: Regime-Conditioned Parametric (`REGIME_PARAMETRIC`)

9.1 Algorithmic Execution

1. **Pre-computation:** For each regime $r \in \mathbf{R}$:

- Estimate conditional mean $\boldsymbol{\mu}_r$ and covariance $\boldsymbol{\Sigma}_r$.
- Perform reactive Cholesky decomposition:

$$\boldsymbol{\Sigma}_r = \mathbf{L}_r \mathbf{L}_r^T$$

- If any regime covariance fails positive-definiteness ($L_{jj}^2 \leq 10^{-10}$), apply reactive Tikhonov diagonal regularization ($\boldsymbol{\Sigma}_{r,\text{stable}} = \boldsymbol{\Sigma}_r + \lambda_r \mathbf{I}$). Log `stabilizationApplied[r] = true`.

2. **Forward Step Loop** for $\tau = 1 \dots H$:

- **Step 2a (Correlated Shock Generation):** Draw $\mathbf{z}_{m,\tau} \sim N(\mathbf{0}, \mathbf{I})$ using deterministic Box-Muller transform. Compute correlated shock: $\{m,\tau\} = \{\mathbf{L}\{R\{m,\tau-1\}\} \mathbf{z}_{m,\tau}\}$. Simulated return: $\{m,\tau\} = \{\boldsymbol{\mu}\{R\{m,\tau-1\}\} + \{\boldsymbol{\varepsilon}\}_{m,\tau}\}$.
- **Step 2b (Portfolio Accounting):** Execute portfolio path update and rebalancing friction.
- **Step 2c (Regime Transition):** Transition regime state via empirical transition matrix $\mathbf{P}$.

10. Regime Initialization Modes

The engine provides five strictly defined starting regime modes:

Starting Mode	Description	Methodology
<code>START_BULL</code>	Path begins in Bull market	$R_{m,0} = \text{BULL}$ for all m
<code>START_BEAR</code>	Path begins in Bear market	$R_{m,0} = \text{BEAR}$ for all m
<code>START_HIGH_VOL</code>	Path begins in High Volatility	$R_{m,0} = \text{HIGH_VOL}$ for all m
<code>START_LOW_VOL</code>	Path begins in Low Volatility	$R_{m,0} = \text{LOW_VOL}$ for all m
<code>START_CURRENT_OBSERVED</code>	Path begins in latest historical state	$R_{m,0} = R_T$ (classification of observation $T = 1,825$)
<code>START_EMPIRICAL_DISTRIBUTION</code>	Sampled from historical occupancy	$R_{m,0} \sim \boldsymbol{\pi} = [T_{\text{BULL}}/T, \dots, T_{\text{LOW_VOL}}/T]$

First-Day Return Convention Invariant:

- **Standard Protocol:** The first simulated return at forward day $\tau = 1$ is generated **directly from the starting regime** R_0 .

- The transition to R_1 occurs at the close of day $\tau = 1$, governing the return generation for day $\tau = 2$. This convention is applied universally across all paths.

11. Regime Persistence & Dwell Time Diagnostics

11.1 Empirical Metrics

1. Regime Occupancy Frequency:

$$\pi_i = \frac{T_i}{T} \in [0, 1], \quad \sum_{i=1}^4 \pi_i = 1.0$$

2. Historical Observed Dwell Time:

- Average duration in consecutive days: $\bar{d}_i = \frac{1}{K_i} \sum_{k=1}^{K_i} d_{i,k}$ (where K_i is number of distinct regime bouts).
- Median duration in consecutive days: $\text{Median}(d_i)$.

3. Self-Transition Probability (Stickiness):

$$P_{ii} = \frac{N_{ii}}{N_i}$$

4. Theoretical Geometric Dwell Time (derived from Markov chain):

$$E[D_i] = \frac{1}{1 - P_{ii}} \quad (\text{valid for } P_{ii} < 1.0)$$

(Note: This represents the theoretical mean duration if transitions were memoryless geometric trials).

5. Switching Frequency:

$$\text{Rate}_{\text{switch}} = \frac{\sum_{i \neq j} N_{ij}}{T - 1}$$

12. Conditional Monte Carlo Outputs & Metric Contracts

For each starting regime configuration, the engine evaluates and returns:

```
export interface RegimeConditionalSimulationResult {
  startingRegime: RegimeType;
  method: 'REGIME_BOOTSTRAP' | 'REGIME_PARAMETRIC';
  simulationCount: number;
  horizonDays: number;
  terminalWealth: PercentileSummary; // P05, P25, P50, P75, P95
  totalReturn: PercentileSummary;    // P05, P25, P50, P75, P95
  maxDrawdown: PercentileSummary;    // P05, P25, P50, P75, P95
  cagr?: PercentileSummary;          // Horizon >= 63 days
  sharpeRatio?: PercentileSummary;   // Horizon >= 63 days
  riskMetrics: {
    lossFrequencyPct: number;        // Paths ending below initial capital
    drawdownExceedance10Pct: number; // Paths with MDD <= -10%
    drawdownExceedance20Pct: number; // Paths with MDD <= -20%
    drawdownExceedance30Pct: number; // Paths with MDD <= -30%
    oneDaySimulatedVaR95: number;
    oneDaySimulatedCVaR95: number;
    horizonSimulatedVaR95: number;
    horizonSimulatedCVaR95: number;
  };
  regimeDiagnostics: {
    averageTransitionsPerPath: number;
    occupancyBreakdownPct: Record<RegimeType, number>;
    terminalRegimeBreakdownPct: Record<RegimeType, number>;
  };
  samplePaths: SamplePath[];        // 50 representative paths
  trajectoryBands: SimulationTrajectoryBands;
  fingerprint: string;
}
```

13. Cross-Regime Comparative Analysis

The engine supports side-by-side comparative matrices evaluating how the starting regime influences simulated outcomes:

REGIME CONDITIONING COMPARATIVE MATRIX				
Starting Regime	P50 Return	P95 MDD	Loss Pct	Median Term\$
BULL	+19.2%	-11.4%	6.2%	\$119,200
BEAR	+12.8%	-18.6%	12.4%	\$112,800
HIGH_VOL	+11.1%	-24.8%	16.8%	\$111,100
LOW_VOL	+16.4%	-9.2%	5.1%	\$116,400

13.1 Neutral Reporting Protocol

- **Prohibition of Value Judgments:** Regimes must **never** be labeled as "best", "worst", "optimal", or "dangerous".
- **Descriptive Standards:**
 - *Compliant:* "Conditioning on START_BEAR produced a simulated P95 maximum drawdown of -18.6% versus -11.4% for START_BULL."
 - *Non-Compliant:* "BEAR is the most dangerous regime and should be avoided."

14. Regime-Conditioned Euler Risk Decomposition

For each regime subset D_r , the engine calculates the conditional Euler risk decomposition:

1. Marginal Risk Contribution:

$$\text{MRC}_{i,r} = \frac{(\boldsymbol{\Sigma}_r \mathbf{w})_i}{\sigma_{p,r}}$$

2. Component Risk Contribution:

$$\text{CRC}_{i,r} = w_i \times \text{MRC}_{i,r}$$

3. Percentage Risk Contribution:

$$\text{PRC}_{i,r} = \frac{\text{CRC}_{i,r}}{\sigma_{p,r}}$$

Euler Identity Invariant:

$$\sum_{i=1}^3 \text{CRC}_{i,r} = \sigma_{p,r}, \quad \sum_{i=1}^3 \text{PRC}_{i,r} = 1.0 \quad (\forall r \in \mathbb{R})$$

This quantifies how asset risk dominance shifts across macro regimes (e.g. Bitcoin PRC expanding from 55% in `LOW_VOL` to 72% in `HIGH_VOL`).

15. Analytical Separation: Historical vs. Unconditional vs. Conditional

The platform strictly differentiates three distinct simulation and analytical layers across all interfaces:

THREE-TIER QUANTITATIVE TAXONOMY		
Tier	Mechanism	Research Question Answered
A. Historical Backtest	Deterministic 2019–2023 Historical Bar Replay	What actually happened over the synchronized dataset?
B. Unconditional Monte Carlo (3.8)	Stationary Resampling Without Regime Switching	What is the outcome spread assuming constant history?
C. Regime-Conditioned Monte Carlo (3.9)	Markov Regime Switching + Conditional Pools	How does outcome dispersion shift based on macro state?

16. Stress Testing Boundary & Interaction

16.1 Decoupled Architecture

- **Phase 3.9 Baseline:** Focuses strictly on stochastic regime-conditioned simulation.
- **Stress Lab Overlay (Optional Advanced Experiment):**
 - If requested, a deterministic Phase 3.2 macro shock (e.g. `COVID_2020` or `TECH_CRASH`) can be applied on Day 1, followed by regime-conditioned forward simulation from Day 2 to *H*.
 - **Terminology Boundary:** Must be labeled *"Stressed Regime Recovery Simulation"*, **NEVER** *"Crisis Forecast"*.

17. High-Throughput Performance & Web Worker Reuse

17.1 Web Worker Reuse Strategy

Phase 3.9 directly reuses the Phase 3.8 Web Worker infrastructure (`monteCarlo.worker.ts` and `monteCarloWorkerClient.ts`).

- **Zero Full-Cube Materialization:** The maximum configuration (50,000 paths $\times$ 1,260 days) evaluates 6.3×10^7 steps. The worker uses streaming histogram reductions, returning:
 - Bounded percentile summaries (P05, P25, P50, P75, P95)
 - 106 sampled trajectory fan coordinates
 - Exactly 50 representative sampled paths
- **Memory Invariant:** Net memory transferred to the main React thread remains < 5 MB, preventing browser freezing or UI thread jank.

18. AI Assistant & Autonomous Quant Research Agent Integration

18.1 Tool Contract: `get_regime_monte_carlo_risk`

```
export const GetRegimeMonteCarloRiskSchema = z.object({
  goldWeight: z.number().min(0).max(1).default(0.4),
  btcWeight: z.number().min(0).max(1).default(0.3),
  nvdaWeight: z.number().min(0).max(1).default(0.3),
  startingRegime: z.enum(['BULL', 'BEAR', 'HIGH_VOL', 'LOW_VOL', 'CURRENT']).default('CURRENT'),
  method: z.enum(['REGIME_BOOTSTRAP', 'REGIME_PARAMETRIC']).default('REGIME_BOOTSTRAP'),
  simulationCount: z.number().int().min(100).max(10000).default(5000),
  horizonDays: z.number().int().min(21).max(504).default(252),
  rebalanceSchedule: z.enum(['BUY_AND_HOLD', 'DAILY', 'MONTHLY', 'QUARTERLY', 'THRESHOLD']).default('MONTHLY'),
  seed: z.number().int().optional(),
});
```

18.2 Tool Contract: `compare_regime_simulations`

```
export const CompareRegimeSimulationsSchema = z.object({
  goldWeight: z.number().min(0).max(1).default(0.4),
  btcWeight: z.number().min(0).max(1).default(0.3),
  nvdaWeight: z.number().min(0).max(1).default(0.3),
  method: z.enum(['REGIME_BOOTSTRAP', 'REGIME_PARAMETRIC']).default('REGIME_BOOTSTRAP'),
  horizonDays: z.number().int().min(21).max(504).default(252),
});
```

18.3 Autonomous Quant Agent Guardrails

- **Capped Experiment Budget:** The agent may execute at most **2 regime simulation runs** per research task.
- **Hypothesis/Evidence Delineation:**
 - *Direct Evidence:* "Simulating 5,000 paths starting from BEAR yielded a P95 maximum drawdown of -21.4% versus -13.2% starting from BULL."
 - *Interpretation:* "The regime transition Markov chain indicates that early BEAR exposure compounds tail drawdown risk before historical mean-reversion occurs."

19. Ghost Mode Insight Categories (Phase 3.1 Extension)

Ghost Mode will support five new deterministic regime insight categories:

- 1. `REGIME_OUTCOME_DIVERGENCE` : Triggered when median terminal wealth differs by $> 20\%$ between starting in `BULL` versus `BEAR`.
- 2. `REGIME_DRAWDOWN_PRESSURE` : Triggered when P95 max drawdown starting in `HIGH_VOL` exceeds $2.0\times$ the historical backtest drawdown.
- 3. `REGIME_TRANSITION_RISK` : Triggered when probability of transitioning into `HIGH_VOL` from `BULL` exceeds 15% .
- 4. `REGIME_PERSISTENCE` : Triggered when a regime self-transition probability $P_{ii} > 0.85$ (indicating high regime inertia).
- 5. `REGIME_RISK_CONTRIBUTION` : Triggered when an asset's conditional PRC expands by > 15 percentage points across regimes.

20. Institutional UI Architecture: Regime Intelligence Lab

Integrated into the Portfolio Cockpit as a dedicated section: **Regime Intelligence Lab**.

REGIME INTELLIGENCE LAB				
[Controls: Starting Regime (BULL BEAR HVOL LVOL) Method Sims Horizon]				
EMPIRICAL TRANSITION MATRIX 4x4 Observed Transition Heatmap - Row-Stochastic Probabilities - Sample Transition Counts (N_{ij})		REGIME OCCUPANCY & DWELL TIME - Occupancy Frequencies (BULL: 42%) - Average Dwell Time (Days) - Self-Transition Stickiness (P_{ii})		
CROSS-REGIME CONDITIONAL COMPARISON				
Starting Mode	P50 Return	P95 MDD	Loss Pct	Median Terminal
START_BULL	+18.4%	-12.1%	6.4%	\$118,400
START_BEAR	+12.1%	-19.8%	13.2%	\$112,100
START_HIGH_VOL	+10.5%	-25.2%	17.4%	\$110,500
START_LOW_VOL	+16.2%	-9.8%	5.2%	\$116,200
CONDITIONAL TRAJECTORY PERCENTILE FAN (STARTING REGIME: HIGH_VOL) P05, P25, P50, P75, P95 Bands with 50 Sample Paths & \$100k Breakeven				
PROVENANCE & AUDIT BAR				
Fingerprint: bx-reg-9a4f21 1,825 Sync Obs Empirical Markov Chain 5k				

20.1 Institutional Visual Design Tokens

- **Canvas / Background:** `#FAF8F4` Warm Ivory
- **Card / Surface:** `#FFFFFF` with `#E5E0D8` Borders
- **Text:** `#1A1917` Graphite (Primary), `#68645E` Muted Graphite
- **Regime Accents:**
 - `BULL` : `#2D7A4C` Institutional Forest Green
 - `BEAR` : `#A83226` Institutional Oxide Red
 - `HIGH_VOL` : `#C87A1E` Institutional Amber
 - `LOW_VOL` : `#255899` Institutional Slate Blue
- **Typography:** Inter (UI / Text) and JetBrains Mono (Matrices, Numbers, Fingerprints)

21. Testing & Verification Requirements (30 Invariants)

The verification suite (`tests/regimeMonteCarlo.test.ts`) must enforce **30 mathematical invariants**:

Test ID	Mathematical Invariant Verified	Assertion / Condition
Test 1	Regime Sequence Alignment	Sequence length strictly equals 1,825 observations.
Test 2	Transition Count Correctness	$\sum_{i,j} N_{ij} = 1,824$ total transitions.
Test 3	Row Normalization	$\sum_j P_{ij} = 1.0$ for all valid originating regimes.
Test 4	Sparse Regime Flagging	Flags warning if regime count < 20 .
Test 5	Zero-Transition Handling	Triggers <code>'INSUFFICIENT_TRANSITION_DATA'</code> on zero outgoing transitions.
Test 6	Return Subset Partitioning	$\sum_r T_r = 1,825$ total return observations.
Test 7	Joint Vector Preservation	Resampled vector matches historical synchronized row.
Test 8	Conditional Covariance Symmetry	Σ_r is symmetric for all regimes.
Test 9	Conditional Cholesky Correctness	$\mathbf{L}_r \mathbf{L}_r^T = \Sigma_r$ to 10^{-8} tolerance.
Test 10	Reactive Regularization	Regularization engages only if pivot $\leq 10^{-10}$.
Test 11	Starting Regime Adherence	Day 1 returns generated strictly from declared starting regime.
Test 12	First-Day Transition Timing	Regime transitions engage at close of Day 1.
Test 13	Transition Simulation Consistency	State transition frequencies converge to P_{ij} on large sample.
Test 14	PRNG Bit-Determinism	Identical seed + config produces bit-identical results.
Test 15	Seed Differentiation	Seed $A \not\equiv$ Seed B produces distinct paths.
Test 16	Terminal Wealth Monotonicity	$P_{05} \leq P_{25} \leq P_{50} \leq P_{75} \leq P_{95}$ strictly holds.
Test 17	Drawdown Non-Positivity	Path drawdowns $D_\tau \leq 0$ strictly.
Test 18	Loss Frequency Boundedness	$f_{\text{loss}} \in [0, 1]$ exactly.
Test 19	Drawdown Exceedance Hierarchy	$P(\text{MDD} \leq -30\%) \leq P(\text{MDD} \leq -20\%) \leq P(\text{MDD} \leq -10\%)$.
Test 20	VaR/CVaR Tail Invariant	Simulated $\text{CVaR}_{95} \geq \text{VaR}_{95}$ under positive loss convention.
Test 21	Occupancy Frequency Conservation	$\sum \pi_i = 1.0$ across all regimes.
Test 22	Average Dwell Time Calculation	Positive finite duration for all observed regimes.
Test 23	Portfolio Weight Simplex	$w_i \geq 0, \sum w_i = 1.0$ strictly preserved.
Test 24	Transaction Cost Accounting	Turnover friction deducted according to Phase 3.7 rules.
Test 25	Three-Tier Output Separation	Historical, Unconditional, and Conditional outputs clearly decoupled.
Test 26	Audit Fingerprint Determinism	SHA-256 fingerprint reproducible across runs.
Test 27	Zero <code>Math.random</code> Usage	Automated scan verifies zero unseeded random calls.
Test 28	No Fabricated Probabilities	Matrix matches empirical historical observation counts.
Test 29	No NaN / Infinity Leaks	All outputs finite numbers across edge cases.
Test 30	Web Worker Parity	Worker async execution yields identical summary results.

22. Edge Cases & Defensive Handling

- Dominated Dataset:** If one regime represents 100% of observations, transition matrix degenerates to an absorbing state ($P_{ii} = 1.0$), matching unconditional simulation.
- Extreme Volatility Blowups** ($|e_{-100\%r^*} \leq -100\%$)*:
 - Asset capital clamps to zero ($V_{i,\tau} = 0$). Total portfolio wealth $V_\tau = \sum V_{i,\tau} \geq 0$ is strictly guaranteed non-negative.
- Short Horizon** ($H < 63$): Annualized CAGR and Sharpe ratios are suppressed or flagged with an institutional note.

4. **Single-Path Edge Case ($M = 1$):** Percentiles collapse cleanly ($P_{05} = P_{50} = P_{95} = X_1$).

23. Institutional Limitations & Disclaimers

The specification enforces the following disclosures:

1. **Offline Demonstration Data:** Metrics and regime labels are derived from the existing offline simulated demonstration dataset (`2019-01-01` to `2023-12-31`). No real historical market feeds are claimed.
2. **Empirical Frequency Baseline:** The regime transition matrix reflects historical transition counts over a finite observation window. It is not an econometric forecast of future regime shifts.
3. **First-Order Markov Assumption:** The model assumes transitions depend only on the current regime state, abstracting away higher-order memory effects or macro cycle phases.
4. **Non-Predictive Research Tool:** Regime-conditioned Monte Carlo simulations are exploratory tools for analyzing portfolio sensitivity to regime shifts. They do not predict future regimes and do not constitute financial, investment, legal, or tax advice.

24. Phase 3.9 Architecture Checklist

- ☒ Existing regime engine reused (Phase 2 `detectRegimes` without threshold redefinition)
- ☒ Regime sequence defined (1,825 synchronized daily observations)
- ☒ Transition matrix defined (Empirical transition counts N_{ij} and frequencies P_{ij})
- ☒ Sparse regime policy defined (`MIN_OBS = 20`, `'INSUFFICIENT_TRANSITION_DATA'` state)
- ☒ Conditional return distributions defined (Partitioned subset D_r with empirical moments)
- ☒ Joint asset preservation defined (Complete multi-asset vectors without asset decoupling)
- ☒ Regime bootstrap defined (Conditional vector resampling with replacement)
- ☒ Regime parametric simulation defined (Conditional μ_r and Σ_r)
- ☒ Cholesky handling defined (Reactive Tikhonov diagonal regularization fallback)
- ☒ Starting regime semantics defined (`START_BULL` , `BEAR` , `HIGH_VOL` , `LOW_VOL` , `CURRENT`)
- ☒ Transition simulation defined (First-day return from R_0 , transition at close of day 1)
- ☒ Regime persistence diagnostics defined (Occupancy π_i , dwell time $\bar{d}_i$, *stickiness* {ii}Pii)
- ☒ Conditional outcome distributions defined (P05, P25, P50, P75, P95)
- ☒ Conditional risk metrics defined (Loss frequency, MDD exceedances, VaR/CVaR)
- ☒ Historical/unconditional/conditional separation defined (Three distinct analytical tiers)
- ☒ Stress boundary defined (Deterministic Phase 3.2 shock overlay decoupled from baseline)
- ☒ Deterministic PRNG reused (32-bit Mulberry32 without `Math.random`)
- ☒ Fingerprint defined (Canonical SHA-256 hash including regime configuration)
- ☒ Web Worker reuse defined (Phase 3.8 worker infrastructure reused, memory bounded)
- ☒ Provenance defined (Complete metadata serialization)
- ☒ Ghost Mode integration defined (5 deterministic regime insight triggers)
- ☒ Research Trail integration defined (Phase 3.6 direct evidence vs interpretation)
- ☒ AI tool contracts defined (`get_regime_monte_carlo_risk` with strict Zod schemas)
- ☒ Quant Agent integration defined (Bounded experiment with capped tool budget)

- ☒ UI architecture defined (Regime Intelligence Lab, Institutional ivory/graphite palette)
- ☒ UI language rules defined (Strictly non-predictive wording protocols)
- ☒ Test strategy defined (30 automated verification test invariants)
- ☒ Edge cases defined (Single regime, sparse data, asset bankruptcy clamping)
- ☒ Security constraints defined (Server-side validation, no code execution, no key exposure)
- ☒ Limitations defined (4 mandatory institutional disclosures)
- ☒ Phase 3.9 boundary defined (Architecture specification only, zero code changes)

End of Phase 3.9 Architecture Specification. Implementation must not proceed until this document has been formally reviewed and explicitly approved.